TAVO: Target-Aware Source Curation for Budgeted Cross-Center Tumor Segmentation

Xiwen Chen[1*] Tong Miao[1*] Yueqi Dong[1]
Divya Rajparia[1,2] Sai Praneeth Karimireddy[1] Ruishan Liu[1†]
[1]University of Southern California, USA [2]IIT Hyderabad, India
xiwenc@usc.edu, miaot@usc.edu, yueqidon@usc.edu
es22btech11013@iith.ac.in, karimire@usc.edu, ruishanl@usc.edu

Abstract

Tumor segmentation at a new clinical center often relies on a small labeled target set and a heterogeneous pool of annotated external cases. Because tumor labels are costly and imaging protocols, patient populations, and tumor appearances vary across centers, using all external cases can increase training cost and induce negative transfer. We introduce ***Target-Aware Valuation Optimization (TAVO)****, a model-external source-curation framework for budgeted cross-center tumor segmentation. Given a source-case budget B, TAVO adaptively combines target similarity, gradient compatibility, source coverage, and diversity by normalizing heterogeneous criterion scores into a common rank space and learning target- and budget-specific simplex weights from held-out target validation utility. The resulting top-B list of external source cases is used by an otherwise standard segmentation pipeline, leaving the architecture, loss, and inference procedure unchanged. We evaluate TAVO on BraTS 2021 and MAMA-MIA, two multi-center MRI tumor-segmentation benchmarks covering eight held-out target centers or domains, with OfficeHome as a compact cross-task control. At a 150-case source budget, TAVO improves target-averaged Dice over the strongest fixed selector by 1.0 points on BraTS and 1.9 points on MAMA-MIA, and over the strongest budget-matched adaptation baseline by 1.0 and 2.2 points, respectively. These results show that target- and budget-adaptive source curation can be more reliable than fixed selection rules and can outperform Target + Full Source while using substantially fewer external cases during final training. Code:* https://github.com/francischen0724/TAVO.

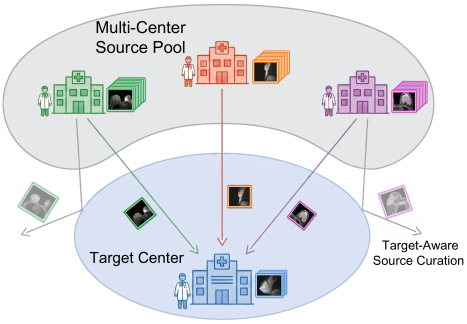


Figure 1. Cross-center source-curation setting. Given limited target support and a heterogeneous external source pool, TAVO curates budgeted source cases before standard segmentation training.

1. Introduction

Tumor segmentation is a particularly challenging setting for cross-center model development. For a new target center, only a small number of labeled cases may be available, because volumetric tumor delineation requires expert voxel-level annotation and tumor appearance can vary substantially in size, shape, location, contrast, and subtype. Meanwhile, annotated external cases may be available from other institutions, clinical cohorts, or public datasets. However, simply pooling all external cases is not necessarily beneficial: scanners, acquisition protocols, treatment cohorts, and patient populations differ across centers, and poorly matched cases can increase computational cost or induce negative transfer [10, 23, 39–41]. Thus, for cross-center tumor segmentation, source composition becomes a first-order design choice: the practical question is not only how to adapt the segmentation model, but also which external cases should be admitted to training for a particular target center. Figure 1 illustrates this cross-center source-curation setting.

Motivated by this setting, we study *budgeted source curation* for cross-center tumor segmentation. We refer to externally annotated cases from other centers or cohorts as

*Equal contributions. †Corresponding author.

source cases. Given a heterogeneous labeled source pool, a small labeled target support set, and a source-case budget B, the goal is to construct a target-adapted training set that improves segmentation at the target center. The budget is case-level: selecting one tumor case counts once toward B and includes all associated slices or patches. Unlike target acquisition, which labels new target cases, we use a fixed labeled target support to curate useful external source cases from an already annotated pool.

We present **Target-Aware Valuation Optimization (TAVO)**, a model-external source-curation framework for cross-center tumor segmentation. Given a target center and source-case budget B, TAVO computes a set of complementary case-level source rankings from a shared warm-up model, covering target similarity, gradient compatibility, source coverage, and diversity. It normalizes these heterogeneous criterion scores into a common rank space and learns target- and budget-specific fusion weights using held-out target validation utility. The fused ranking is converted into a deterministic top-B source-case manifest and consumed by an otherwise standard segmentation training pipeline. TAVO changes neither the segmentation architecture nor the task loss and introduces no inference-time overhead. Its output is a curated training set rather than a new deployment model, making it orthogonal to model-centric adaptation.

We evaluate TAVO on two multicenter MRI tumor-segmentation benchmarks: multimodal brain tumor segmentation from BraTS 2021 and breast tumor segmentation from MAMA-MIA [1, 6]. Together, these experiments cover two anatomies, two MRI protocols, eight held-out target centers, multiple case-level source budgets, and two segmentation backbones. We compare TAVO with random selection, eight fixed source-selection criteria, budget-matched model-centric adaptation baselines, and non-budgeted full-source training. The results show that the strongest fixed criterion changes across targets and budgets, while TAVO provides consistently strong target-averaged performance and can exceed Target + Full Source while using substantially fewer external cases during final training. We retain OfficeHome as a compact cross-task control to test whether the same rank-fusion interface transfers beyond MRI tumor segmentation [33].

Finally, we summarize our contributions as follows:

- We formulate budgeted external-case curation for cross-center tumor segmentation, where limited labeled target support guides which externally annotated tumor cases are admitted to downstream training.
- We introduce TAVO, a model-external data interface that combines target similarity, gradient compatibility, source coverage, and diversity into a target- and budget-specific source-case manifest while leaving the segmentation architecture, task loss, and inference procedure unchanged.
- We provide a comparative assessment on two multicenter MRI tumor-segmentation benchmarks, covering eight held-out target centers or domains, multiple case budgets, eight source-selection criteria, budget-matched adaptation baselines, and two segmentation backbones.

2. Related Work

Cross-center tumor segmentation. Tumor segmentation benchmarks such as BraTS and MAMA-MIA have enabled systematic evaluation across institutions, protocols, and patient cohorts [1, 6]. Standardized segmentation pipelines such as nnU-Net provide strong task-specific baselines, but cross-center variation remains a persistent challenge when a model trained on pooled external data is applied to a new target center [10, 12, 39]. Most work in this area improves architectures, preprocessing, augmentation, or adaptation objectives while assuming that the source training set is fixed. Our focus is orthogonal: we ask which externally annotated tumor cases should be selected under a case-level source budget before standard downstream training.

Data subset selection and valuation. Data subset selection seeks compact training sets that preserve useful learning signals while reducing training cost. Submodular objectives and geometric coresets prioritize coverage, representativeness, or diversity [29, 36], while optimization-aware methods select examples based on training dynamics or gradient approximation. CRAIG approximates the full-data gradient, GradMatch matches subset gradients to a training or validation objective, and GLISTER optimizes held-out utility through bilevel subset selection [14, 15, 22]. Related data-valuation methods estimate scalar example contributions using influence functions, Data Shapley, or training-trajectory attribution [7, 16, 25]. These methods provide useful notions of coverage, optimization relevance, and example value, but they do not by themselves specify how heterogeneous criteria should be combined for a particular target center and source budget.

Target-aware data selection. Several methods use held-out or target data to adapt data importance. Meta-reweighting and Meta-Weight-Net learn sample weights from clean meta-data [26, 30], while DVRL and DDS learn data-value estimators from downstream feedback [35, 38]. Under domain shift, ORIENT uses target-query submodular mutual information to balance source–target relevance and diversity [13]; Selectivity prunes source data using source–target label and feature mappings [41]; and LESS identifies examples through target-relevant gradient similarity [37]. These methods show that source utility can depend on target evidence, validation feedback, or an available budget. However, they typically instantiate a single predefined selection objective, a single learned valuation mechanism, or a fixed weighting

policy. TAVO instead optimizes a target- and budget-specific simplex combination of heterogeneous criteria, retaining single-criterion selectors as simplex corners and inducing a hard top-B source subset.

Model-centric domain adaptation and multi-source transfer. Domain adaptation addresses source–target shift primarily by modifying the learning objective or representation space. Discrepancy-based methods such as DAN/MMD reduce distribution mismatch with kernel distances [18], and CORAL aligns second-order feature statistics [32]. Adversarial methods such as DANN and CDAN instead learn domain-invariant representations [5, 19]. For segmentation, ADVENT performs adversarial entropy minimization over structured outputs, and medical adaptation methods such as SE-ASA combine selective entropy constraints with semantic alignment [4, 34]. Few-shot or semi-supervised adaptation methods such as MME and CLDA use limited target labels, often together with unlabeled target data [28, 31], and multi-source methods such as MDAN and M3SDA model multiple source domains through adversarial bounds or moment matching [24, 42]. These methods adapt the learner or aggregate domains, whereas TAVO addresses the sample-level source-composition decision under an explicit cardinality budget.

3. Methods

Figure 2 gives an overview of TAVO. We first formalize budgeted source curation for cross-center tumor segmentation, then describe how complementary source rankings are aligned and fused, and finally present the validation-guided multi-fidelity search used to select target- and budget-specific fusion weights. After the top-B source-case manifest is fixed, final training follows the same downstream architecture, task loss, checkpointing rule, and inference procedure as the comparison methods.

3.1. Problem Setting

Let $\mathcal{S} = \bigcup_{d=1}^{D} \mathcal{S}_d = \{u_i = (x_i, y_i)\}_{i=1}^{N}$ denote a labeled source pool collected from D external domains. In tumor segmentation, each source example u_i is a complete patient case or volume; selecting it includes all associated slices used by the downstream learner. The labeled target data are split into disjoint training, validation, and test sets, $\mathcal{T}_{\mathrm{tr}}^{\ell}$, $\mathcal{T}_{\mathrm{val}}^{\ell}$, and $\mathcal{T}_{\mathrm{te}}$. The training support provides supervision and target evidence for target-conditioned criteria. The validation support provides the outer-loop utility for fusion selection and checkpoint selection, but is not used to compute criterion scores or train final models. The test set is reserved for final evaluation.

Let θ_{init} denote the common task-specific model initialization. For a candidate source subset $\mathcal{A} \subseteq \mathcal{S}$, define

$$\theta(\mathcal{A}, \xi) = \mathrm{Train}_{\mathrm{full}}\left(\theta_{\mathrm{init}}; \mathcal{A} \cup \mathcal{T}_{\mathrm{tr}}^{\ell}, \xi\right), \tag{1}$$

where ξ represents training randomness. Given a source-cardinality budget $B \leq N$, the ideal subset would maximize held-out target validation utility,

$$\mathcal{A}_B^{\mathrm{opt}} \in \arg\max_{\substack{\mathcal{A} \subseteq \mathcal{S} \\ |\mathcal{A}| = B}} \mathbb{E}_{\xi}\left[Q\left(f_{\theta(\mathcal{A},\xi)}; \mathcal{T}_{\mathrm{val}}^{\ell}\right)\right], \tag{2}$$

where Q denotes the target-validation metric evaluated on $\mathcal{T}_{\mathrm{val}}^{\ell}$. Exhaustively evaluating the $\binom{N}{B}$ possible subsets is infeasible, so TAVO restricts the search to a structured family of subsets induced by a low-dimensional fusion of source-valuation criteria. The source-case budget B constrains the number of source examples used in candidate and final training.

3.2. Target-Aware Multi-Criteria Rank Fusion

TAVO combines M source-valuation criteria through a common ranking interface. In our segmentation experiments, we use $M = 8$ complementary criteria organized into two families. The target-conditioned criteria are RDS [20, 27], LESS [37], ORIENT [13], and GradMatch [14]. The source-structural criteria are CRAIG [22], KMeans [17], KCenter [29], and Diversity [8]. Together, these criteria provide complementary views of target relevance, gradient compatibility, source coverage, and diversity, allowing TAVO to trade off target-conditioned and source-structural evidence rather than relying on one fixed selection rule.

To compute the feature and gradient evidence required by these criteria, we construct a fixed criterion warm-up model $f_{\theta_{\mathrm{warm}}}$. Starting from θ_{init}, the warm-up model is briefly trained on $\mathcal{S} \cup \mathcal{T}_{\mathrm{tr}}^{\ell}$ and is then used only to extract representations and gradients for source valuation. Candidate and final models are reinitialized from θ_{init}. This full-pool pass is used only for valuation evidence; the budget B constrains the source cases used in candidate and final training. The warm-up schedule and criterion-specific implementations are provided in Supplementary Secs. A.1–A.2.

Each criterion produces either a per-example utility score or an ordered prefix of selected source cases. Score-based criteria are sorted by descending utility, while subset-producing criteria retain their native selection order. Examples outside a returned prefix receive a common tail rank below the listed examples. Criteria whose native output depends on the requested subset size are recomputed separately for each budget B.

Because native utilities are not calibrated across criteria, TAVO fuses ranks rather than raw scores. Let $\rho_{i,B}^{(m)} \in [1, N]$ denote the descending rank assigned to source example i by criterion m under budget B, where rank 1 indicates the

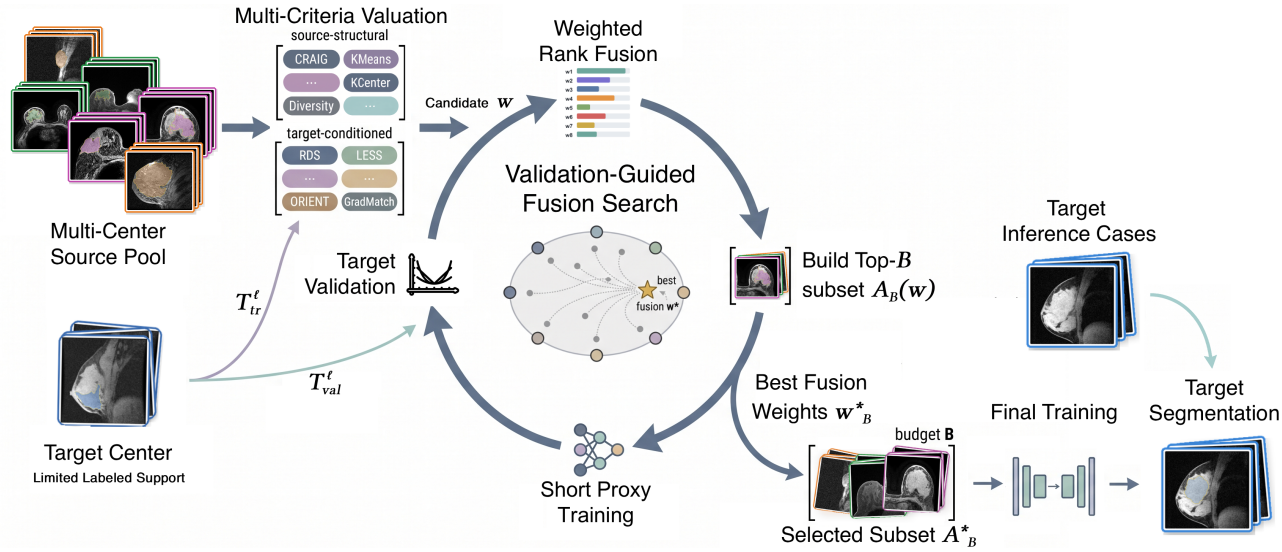


Figure 2. Overview of TAVO. Given a heterogeneous labeled source pool and limited labeled target support, TAVO computes complementary case-level source rankings from target-conditioned and source-structural criteria. These rankings are aligned in a common rank space and combined by candidate fusion vectors. Each candidate w induces a top-B source subset $A_B(w)$, which is assessed through short proxy training and held-out target validation. The selected fusion $w^{\star}_B$ is frozen, and the resulting source-case manifest $A^{\star}_B$ is used together with the labeled target training support for final tumor-segmentation training.

highest estimated utility. We define

$$v^{(m)}_{i,B} = \frac{N - \rho^{(m)}_{i,B}}{N-1}, \qquad v^{(m)}_{i,B} \in [0,1]. \tag{3}$$

This transformation places all criteria on a common scale while preserving their preference order. Let $\mathbf{v}_{i,B} = [v^{(1)}_{i,B}, \dots, v^{(M)}_{i,B}]^{\top}$ denote the normalized valuation vector of source example i, and let $\mathbf{V}_B \in [0,1]^{N \times M}$ be the aligned matrix obtained by stacking these vectors.

For each target domain and source budget, TAVO optimizes a nonnegative fusion vector on the probability simplex,

$$\mathbf{w} \in \Delta^{M-1} = \left\{ \mathbf{w} \in \mathbb{R}^M : w_m \geq 0, \ \sum_{m=1}^{M} w_m = 1 \right\}. \tag{4}$$

The fused score for source example i is

$$s_{i,B}(\mathbf{w}) = \mathbf{w}^{\top} \mathbf{v}_{i,B} = \sum_{m=1}^{M} w_m v^{(m)}_{i,B}. \tag{5}$$

For tumor segmentation, TAVO selects a global top-B subset of source cases:

$$\mathcal{A}_B(\mathbf{w}) = \mathrm{Top}_B \left(\{(u_i, s_{i,B}(\mathbf{w}))\}_{i=1}^{N} \right). \tag{6}$$

Simplex vertices correspond to individual criteria, while interior points fuse multiple criteria. Thus, TAVO learns a fused scoring rule rather than imposing predefined quotas over criterion families or source domains.

3.3. Validation-Guided Multi-Fidelity Search

The fused scores in Eq. (5), together with the top-B operator in Eq. (6), define a candidate source subset for any fusion vector $\mathbf{w}$. However, the usefulness of this subset is determined by downstream target performance, not by the criterion matrix alone. TAVO therefore scores fusion vectors through proxy training on their induced subsets and evaluation on $\mathcal{T}^{\ell}_{\mathrm{val}}$. Since full training for every candidate is expensive, we use a fixed multi-fidelity protocol: (i) a coarse stage evaluates many candidate fusion weights with short proxy-training runs, and (ii) a refinement stage re-evaluates a small set of promising candidates with longer proxy-training runs.

For fusion vector $\mathbf{w}$, training seed ξ, and evaluated checkpoint k, let

$$\theta_k(\mathbf{w}, \xi) = \mathrm{Train}_k \left(\theta_{\mathrm{init}}; \mathcal{A}_B(\mathbf{w}) \cup \mathcal{T}^{\ell}_{\mathrm{tr}}, \xi \right). \tag{7}$$

For a proxy-training length J, candidate fitness is

$$\widehat{F}_J(\mathbf{w}; \Xi) = \mathrm{median}_{\xi \in \Xi} \left[\max_{k \in \mathcal{K}_J} Q\left(f_{\theta_k(\mathbf{w},\xi)}; \mathcal{T}^{\ell}_{\mathrm{val}} \right) \right], \tag{8}$$

where $\mathcal{K}_J$ is the fixed set of checkpoints evaluated within the J-step proxy-training run, Ξ is a predetermined set of training seeds, and Q is the target-validation segmentation metric. We use J_{c} for coarse exploration and a longer horizon $J_{\mathrm{r}} > J_{\mathrm{c}}$ to refine the top R coarse-stage configurations from scratch.

Algorithm 1 TAVO search for one target–budget setting

Require: Aligned criterion matrix $\mathbf{V}_B$, budget B, subset operator $\mathcal{A}_B$
Require: Proxy-training horizons $J_c < J_r$, seed sets Ξ_c, Ξ_r
Require: CMA-ES parameters $G, \lambda, \mu, \sigma_0$, refinement size R
Ensure: Fusion weights $\mathbf{w}_B^\star$ and selected subset $\mathcal{A}_B^\star$

$\mathcal{H} \leftarrow \emptyset$
$\mathcal{W}_0 \leftarrow \{\mathbf{e}_m\}_{m=1}^M \cup \{M^{-1}\mathbf{1}\}$
for all $\mathbf{w} \in \mathcal{W}_0$ **do**
  $q \leftarrow \widehat{F}_{J_c}(\mathbf{w}; \Xi_c)$
  Add $(\mathbf{w}, q)$ to $\mathcal{H}$
end for
$\Omega \leftarrow \textsc{InitCMA}(M^{-1}\mathbf{1}, \sigma_0)$
for $g = 1, \ldots, G$ **do**
  $\{(\mathbf{z}_j, \mathbf{w}_j)\}_{j=1}^\lambda \leftarrow \textsc{SampleAndMapCMA}(\Omega, \lambda)$
  for $j = 1, \ldots, \lambda$ **do**
    $q_j \leftarrow \widehat{F}_{J_c}(\mathbf{w}_j; \Xi_c)$
    Add $(\mathbf{w}_j, q_j)$ to $\mathcal{H}$
  end for
  $\Omega \leftarrow \textsc{UpdateCMA}(\Omega, \{(\mathbf{z}_j, q_j)\}_{j=1}^\lambda, \mu)$
end for
$\mathcal{C}_R \leftarrow \textsc{TopDistinctWeights}(\mathcal{H}, R)$
for all $\mathbf{w} \in \mathcal{C}_R$ **do**
  Refine: $\widetilde{q}(\mathbf{w}) \leftarrow \widehat{F}_{J_r}(\mathbf{w}; \Xi_r)$
end for
$\mathbf{w}_B^\star \leftarrow \arg\max_{\mathbf{w} \in \mathcal{C}_R} \widetilde{q}(\mathbf{w})$
$\mathcal{A}_B^\star \leftarrow \mathcal{A}_B(\mathbf{w}_B^\star)$
return $\mathbf{w}_B^\star, \mathcal{A}_B^\star$

The objective in Eq. (8) is non-smooth because of hard top-B source selection and noisy because of stochastic network training. We therefore use CMA-ES [11], a derivative-free population-based optimizer, to propose fusion weights. Each raw CMA-ES sample $\mathbf{z} \in \mathbb{R}^M$ is mapped to the simplex by nonnegative normalization, $\mathbf{w} = [\mathbf{z}]_+ / \mathbf{1}^\top [\mathbf{z}]_+$, with the uniform vector used when the denominator is zero. We also explicitly evaluate all M simplex corners and the uniform fusion vector, so every individual criterion and the equal-weight mixture are included in the search. Algorithm 1 summarizes the coarse-to-refinement search for one target–budget setting.

Here, $\textsc{TopDistinctWeights}(\mathcal{H}, R)$ returns the R distinct fusion vectors with the highest coarse-fidelity scores. Every refinement evaluation is reinitialized from θ_{init}, rather than continued from its coarse-training trajectory. Exact training horizons, checkpoint schedules, seed sets, CMA-ES parameters, and proxy-training horizons are provided in Supplementary Sec. A.3. All search settings are fixed before test evaluation.

3.4. Final Training and Computational Interface

After search, TAVO freezes the selected source-case manifest $\mathcal{A}_B^\star$, reinitializes the task model from θ_{init}, and trains on $\mathcal{A}_B^\star \cup \mathcal{T}_{\text{tr}}^\ell$ using the same full-training protocol as the comparison methods. During final training, $\mathcal{T}_{\text{val}}^\ell$ is used only for checkpoint selection, and $\mathcal{T}_{\text{te}}$ is evaluated only after the source subset, fusion weights, and checkpoint have all been fixed.

TAVO changes neither the downstream architecture nor the task loss, and therefore introduces no inference-time overhead. Given cached criterion rankings, producing a fused source ordering requires $\mathcal{O}(NM + N \log N)$ operations in our implementation. The measured costs of criterion warm-up, evidence extraction, criterion scoring, multi-fidelity search, and final training are reported in Sec. 4.3 and Supplementary Sec. A.4.

4. Experiments

4.1. Experimental Setup

Benchmarks, protocols, and metrics. We evaluate TAVO primarily on two cross-center medical image segmentation benchmarks, with a compact natural-image classification control experiment reported in Sec. 4.6. BraTS 2021 provides 1,251 annotated multimodal brain MRI training cases, with annotations for enhancing tumor (ET), tumor core (TC), and whole tumor (WT) [1, 2, 21]. MAMA-MIA contains 1,506 pre-treatment breast DCE-MRI cases from four centers: NACT, I-SPY1, I-SPY2, and Duke [6]. OfficeHome contains 15,588 images from 65 categories across Art, Clipart, Product, and Real World [33]. We use dataset-provided center labels and official case-mapping files to define domains, and use fixed split manifests for all reported experiments.

For segmentation, we use leave-one-domain-out evaluation: one center/domain is held out as the target, and all remaining domains form the labeled source pool. Target cases are split into disjoint training, validation, and test sets, $\mathcal{T}_{\text{tr}}$, $\mathcal{T}_{\text{val}}$, and $\mathcal{T}_{\text{te}}$. The training split supervises candidate and final training and provides target evidence for target-conditioned criteria; the validation split selects TAVO fusion weights and model checkpoints; and the test split is accessed only for final reporting.

For BraTS, we report 3D case-level Macro Dice averaged over ET, TC, and WT. For MAMA-MIA, we report foreground Dice. For both benchmarks, we evaluate source budgets $B \in \{50, 150, 250, 500\}$ cases. Each setting is repeated with three final-training seeds, and dataset-level summaries are unweighted averages over the four held-out target domains. Exact target split sizes, source-pool sizes, and source-budget percentages are provided in Supplementary Sec. B.1.

Baselines and fairness. We compare TAVO with three groups of methods: (i) non-budgeted references, including Target-only, Full Source-only, and Target + Full Source; (ii) budget-matched source-curation baselines, including

Random and the eight fixed single-criterion selectors spanning the target-conditioned and source-structural criteria described in Sec. 3.2; and (iii) model-centric adaptation baselines for segmentation, including DANN, MMD, ADVENT, and SE-ASA [4, 5, 18, 34]. All budgeted selectors and budget-matched adaptation baselines use exactly B source examples together with the same labeled target training support. Budget-matched adaptation baselines are trained on a random B-case source subset, so adaptation is evaluated under the same source-cardinality constraint as the selection methods. Full-source references are reported only as non-budgeted references.

Implementation and reporting. Within each benchmark, all methods share the same preprocessing, backbone, optimizer schedule, training length, and checkpoint-selection rule. BraTS and MAMA-MIA use nnU-Net preprocessing, 2D slice-level training, and reconstructed 3D case-level evaluation [12]. BraTS uses EfficientViT-Seg-L1 initialized from ADE20K-pretrained weights [3, 43], which provides an efficient pretrained segmentation backbone for repeated subset training and valuation search; MAMA-MIA uses the nnUNet v2 2D configuration with three DCE channels, allowing us to evaluate TAVO on a standardized medical segmentation pipeline [12]. Final training uses 20 epochs for BraTS and 100 epochs for MAMA-MIA.

For each held-out target, we train one early Target + Full Source warm-up model and use its frozen checkpoint only to cache representations and gradients for source valuation. Candidate and final models are reinitialized from the same task-specific initialization and trained from scratch. Search hyperparameters and proxy-training lengths are fixed before test evaluation and reported in Supplementary Sec. A.3. The source budget constrains candidate and final training; the one-time full-pool warm-up, evidence extraction, and criterion scoring costs are reported in Sec. 4.3. Additional implementation and metric details are provided in Supplementary Secs. B.2–B.3.

4.2. Main Results under Matched Source Budgets

Overall performance across budgets. Figure 3 summarizes target-averaged segmentation performance under matched source budgets. Across BraTS 2021 and MAMA-MIA, TAVO remains among the strongest methods as the source budget varies, despite using a single adaptive fusion strategy rather than a fixed target-conditioned, source-structural, or adaptation objective. On both medical benchmarks, TAVO also outperforms the Target + Full Source reference in most budget settings while using substantially fewer external cases during final training. This shows that cross-center source data are not uniformly beneficial: under domain shift, indiscriminately adding all available source cases can be suboptimal, and the composition of the source set can matter as much as its size. Complete per-target results across all source budgets are provided in Supplementary Sec. C.1.

Representative tumor segmentation results. Table 1 reports per-target results at the representative 150-case source budget. TAVO achieves 77.9 Macro Dice on BraTS and 71.4 foreground Dice on MAMA-MIA, improving over Random by 2.3/2.2 points, the strongest fixed selector by 1.0/1.9 points, and the strongest budget-matched adaptation baseline by 1.0/2.2 points, respectively. The gain is not driven by a single favorable target: TAVO ranks first on five of the eight tumor-segmentation targets and second on the remaining three, while the best competing fixed selector changes across centers. This supports the premise that source utility is target-dependent and cannot be captured reliably by one fixed rule. TAVO also exceeds the non-budgeted Target + Full Source reference on all eight targets, improving the target average by 2.9 points on BraTS and 2.0 points on MAMA-MIA, suggesting that under cross-center shift, source composition can matter more than simply using all available external cases. Qualitative examples in Figure 4 provide representative contour-level comparisons. Additional paired significance analyses and stronger-counterpart evaluations are provided in Supplementary Secs. C.4–C.5 and further support the headline comparisons.

4.3. Computational Cost and Practical Trade-offs

Table 2 reports TAVO's one-time development cost at $B = 150$. The source budget constrains candidate and final training, while warm-up and evidence extraction access the full source pool once per target. Total cost is $1.19\times$ /$1.15\times$ one Target + Full Source run on BraTS/MAMA-MIA, while remaining below a full-fidelity eight-selector sweep (61.6/56.8 GPU-hours). This overhead is confined to development: once the source manifest is fixed, TAVO adds no deployment- or inference-time cost. Search-budget sensitivity is reported in Supplementary Sec. C.6.

4.4. Target- and Budget-Specific Fusion Behavior

Learned fusion weights. Figure 5 shows that TAVO does not learn a single target-invariant weighting policy. Both the target-conditioned weight mass and the effective number of active criteria, $N_{\text{eff}} = 1/\sum_{m=1}^{M} w_m^2$, vary across held-out centers and source budgets, indicating that TAVO adapts the trade-off between target relevance and source-structural coverage to each transfer setting. At the representative 150-case source budget, the criterion-level profiles also differ across targets, consistent with Table 1, where the strongest fixed selector changes by center. We therefore interpret the learned weights as descriptive fusion profiles rather than causal estimates of criterion importance.

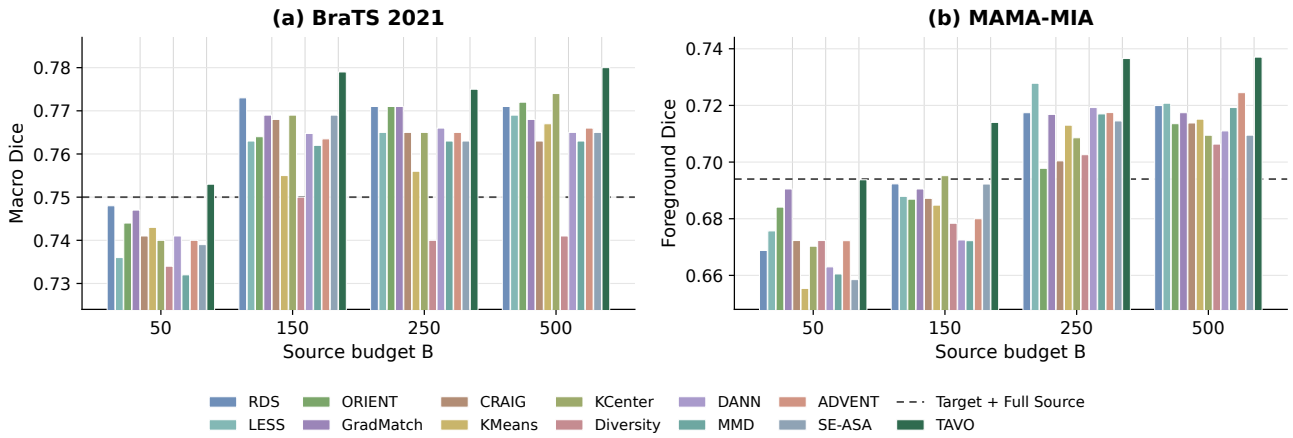


Figure 3. Target-averaged performance across source budgets on BraTS 2021 and MAMA-MIA. Each method is evaluated under matched source budgets; the dashed line denotes the non-budgeted Target + Full Source reference.

Table 1. Per-target segmentation results at the source budget ($B = 150$). The top block contains non-budgeted references; LGG and GBM denote TCGA-LGG and TCGA-GBM; Boldface and underlining indicate the best and second-best budget-matched results, respectively.

Category	Method	BraTS 2021 Macro Dice ↑					MAMA-MIA Foreground Dice ↑				
		C4	C5	LGG	GBM	Avg.	NACT	I-SPY1	Duke	I-SPY2	Avg.
Non-budgeted refs.	Target-only	70.7±0.3	77.7±0.1	56.5±0.3	82.4±0.2	71.8	53.6±0.7	66.1±0.5	56.3±0.1	74.7±0.2	62.7
	Full Source-only	76.0±0.6	73.4±0.2	59.5±0.7	79.0±0.4	72.0	71.5±0.4	69.6±0.3	56.0±0.4	62.5±0.8	64.9
	Target + Full Source	76.1±0.5	79.1±0.3	61.6±0.4	83.2±0.2	75.0	72.1±0.5	70.6±0.2	59.4±0.5	75.4±0.6	69.4
Random	Random	75.9±1.1	82.4±0.8	60.9±0.6	83.4±0.3	75.6	67.8±0.5	69.2±0.6	63.5±0.3	76.3±0.8	69.2
Target-conditioned	RDS [20, 27]	76.5±0.1	81.4±0.2	**63.8**±1.8	84.6±0.2	76.6	70.3±0.8	65.8±0.2	**64.4**±0.4	76.4±0.4	69.2
	LESS [37]	77.9±1.0	80.8±0.4	62.6±0.6	83.8±0.2	76.3	69.9±0.3	67.8±0.2	62.4±0.3	75.1±0.4	68.8
	ORIENT [13]	76.9±1.3	82.7±0.4	61.7±0.4	84.4±0.0	76.4	67.4±0.3	71.1±0.2	60.0±0.7	76.2±0.5	68.7
	GradMatch [14]	78.5±1.7	82.3±0.1	62.4±1.9	84.5±0.4	76.9	68.6±0.5	68.9±0.3	62.5±0.5	76.2±0.6	69.1
Source-structural	CRAIG [22]	78.8±1.3	82.2±0.4	61.8±2.0	84.2±0.0	76.8	69.6±0.2	67.7±0.4	61.9±0.4	75.7±0.2	68.7
	KMeans [17]	77.5±1.3	82.2±0.3	58.0±1.4	84.3±0.1	75.5	66.0±1.0	69.3±0.4	62.4±1.0	76.2±0.5	68.5
	KCenter [29]	78.7±1.1	82.4±0.1	62.4±0.6	84.0±0.2	76.8	71.0±0.2	68.1±0.2	63.2±0.5	75.8±1.1	69.5
	Diversity [8]	79.0±1.7	80.2±0.1	57.5±1.1	83.2±0.2	75.0	65.9±0.9	68.7±0.3	60.6±1.3	76.2±0.3	67.8
Budget-matched DA	DANN [5]	78.3±2.0	82.2±0.0	61.0±2.4	84.4±0.3	76.5	67.0±1.1	66.2±0.1	61.6±0.2	74.2±0.6	67.3
	MMD [18]	78.2±0.7	81.7±0.9	60.7±1.5	84.2±0.4	76.2	68.1±0.9	64.3±0.3	62.7±0.6	73.8±0.4	67.2
	ADVENT [34]	77.6±2.2	82.1±0.0	61.2±0.4	84.5±0.1	76.4	67.5±0.4	65.1±0.2	62.9±1.0	76.5±0.4	68.0
	SE-ASA [4]	**80.2**±0.6	82.6±0.2	60.9±0.7	83.9±0.1	76.9	69.4±0.3	68.9±0.5	63.2±0.6	75.4±0.1	69.2
Ours	**TAVO**	79.9±0.6	**83.0**±0.5	63.7±0.9	**84.9**±0.1	**77.9**	**72.9**±0.3	**71.2**±0.2	64.2±0.5	**77.3**±0.1	**71.4**

Table 2. Measured GPU-hours for representative tumor-segmentation settings at budget ($B = 150$) on an NVIDIA A40 GPU. Costs are reported for one final-training seed.

Cost component	BraTS	MAMA-MIA
Criterion warm-up checkpoint	7.8	6.9
Selection evidence extraction	5.6	4.1
Criterion scoring	0.2	0.2
Multi-fidelity search	18.5	15.9
Curation overhead	32.1	27.1
Final training	6.0	5.7
TAVO end-to-end	38.1	32.8
Target + Full Source training	32.0	28.6
End-to-end / full-source training	1.19×	1.15×

Table 3. Ablation at the representative source budget used for each benchmark. Values are averaged over the four target domains.

Variant	BraTS	MAMA-MIA
Source-structural only	76.2	69.2
Target-conditioned only	76.3	70.3
Uniform fusion	76.1	69.4
Val.-selected single	75.9	68.5
Dirichlet random search	75.2	67.9
Shuffled-ranking CMA-ES	76.0	69.5
Full TAVO	**77.9**	**71.4**

4.5. Ablation Studies

Table 3 summarizes TAVO's ablations. Full TAVO exceeds the best single-family, Uniform-fusion, and Val.-selected-

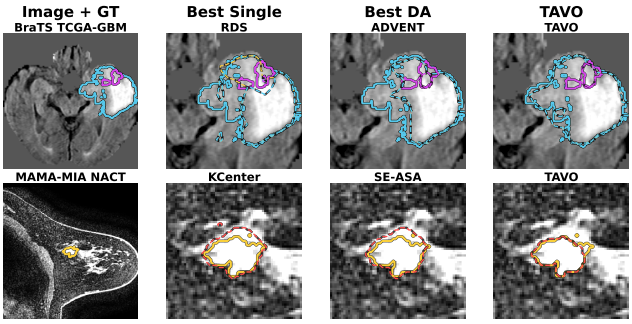


Figure 4. Qualitative segmentation examples at the 150-case source budget ($B = 150$). Solid contours denote ground truth and dashed contours denote predictions. For BraTS, WT, TC, and ET are shown in yellow, cyan, and magenta, respectively. For MAMA-MIA, ground-truth and predicted tumor contours are shown in yellow and red, respectively.

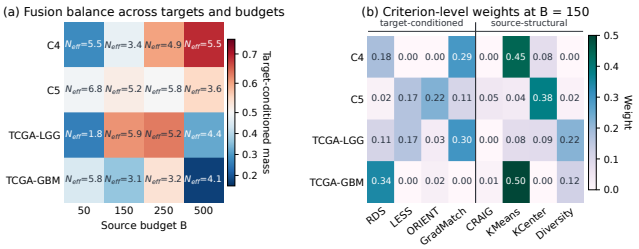


Figure 5. Target- and budget-specific fusion behavior on BraTS.

single variants by 1.6/1.1, 1.8/2.0, and 2.0/2.9 points on BraTS/MAMA-MIA, respectively, and outperforms Dirichlet random search. Shuffling criterion rankings under the same CMA-ES search reduces performance to 76.0/69.5, showing that both informative rankings and adaptive search contribute (see Supplementary Sec. C.3 for per-target results).

4.6. Cross-Task Control on OfficeHome

To test whether the same source-curation interface transfers beyond medical segmentation, we include a compact OfficeHome classification control experiment [33]. Each task pools three domains as the source pool and holds out the remaining domain as the target, with three target-training and two validation images per class; source budgets select b source images per class. Since OfficeHome has 65 categories, the total source budget is $B = 65b$. All methods use an ImageNet-pretrained ResNet-50 and are trained for 30 epochs. TAVO uses the same rank-fusion interface as in segmentation, but replaces the segmentation criterion library with classification counterparts: MomentMatching [9, 18], GradMatch [14], GLISTER [15], and ORIENT [13] as target-conditioned criteria, and CRAIG [22], KMeans [17], KCenter [29], and Facility Location [36] as source-structural criteria. We compare against Random, fixed selectors, and budget-matched

Table 4. OfficeHome accuracy at $b = 15$ selected source images per class, corresponding to $B = 975$ total selected source images. The top block contains non-budgeted references; bold and underline indicate the best and second-best budget-matched results, respectively.

Method	Art	Clipart	Product	Real	Avg.
Target-only	43.9±1.5	26.3±0.4	56.6±1.8	51.2±1.6	44.5
Source-only Full	60.3±1.1	40.3±0.8	75.2±0.2	75.5±0.7	62.8
Target + Full Source	63.3±1.0	49.8±0.4	78.7±0.5	77.0±0.5	67.2
Random	56.6±1.3	44.0±0.9	71.1±0.3	68.3±1.0	60.0
KMeans [17]	57.7±0.3	44.7±1.1	72.3±1.5	68.0±0.5	60.7
KCenter [29]	49.5±0.7	40.2±0.2	67.7±1.0	61.6±1.6	54.7
Facility Location [36]	58.0±1.2	43.9±0.8	72.4±0.8	68.9±0.5	60.8
GradMatch [14]	58.6±1.2	44.3±0.9	73.6±0.6	66.4±1.3	60.7
CRAIG [22]	55.4±0.5	44.6±0.3	72.7±0.7	69.2±0.4	60.5
GLISTER [15]	59.7±1.0	43.9±1.3	74.9±0.7	68.1±0.7	61.7
ORIENT [13]	**60.6**±0.4	46.1±1.2	73.5±0.4	68.4±0.4	62.2
MomentMatching [9, 18]	58.2±1.1	44.1±0.6	73.4±0.1	66.8±0.9	60.6
DANN [5]	57.9±0.2	41.6±0.3	70.9±0.9	68.9±0.8	59.8
MMD [18]	58.5±0.6	41.2±0.7	70.6±0.5	68.7±0.4	59.8
CORAL [32]	57.1±0.3	41.6±0.6	70.7±0.2	68.9±0.6	59.6
CDAN [19]	58.3±0.5	42.0±0.5	70.9±1.1	69.1±0.4	60.1
TAVO	59.7±1.0	**46.5**±0.3	**75.2**±0.7	**69.5**±0.7	**62.7**

classification adaptation baselines DANN [5], MMD [18], CORAL [32], and CDAN [19]. At $b = 15$ in Table 4, TAVO achieves the strongest budget-matched average accuracy (62.7%), outperforming Random, the strongest fixed selector, and the strongest budget-matched adaptation baseline by 2.7/0.5/2.6 points, respectively. These results provide supporting evidence that TAVO is not tied to a particular segmentation architecture, medical imaging modality, or evaluation metric.

5. Conclusion

We introduced TAVO, a model-external source-curation workflow for budgeted cross-center tumor segmentation. TAVO rank-normalizes complementary valuation signals and learns target- and budget-specific fusion weights from held-out target validation, producing a source-case manifest for an otherwise unchanged training and inference pipeline.

Across BraTS 2021 and MAMA-MIA, TAVO consistently improves over random selection and strong fixed selectors under matched source budgets, and in medical settings exceeds Target + Full Source while using fewer external cases during final training. OfficeHome further supports transfer beyond MRI segmentation. These results highlight source composition as a first-class model-development decision under domain shift. TAVO requires labeled target validation, a full-pool warm-up, and target–budget-specific search; future work should reduce these costs, study robustness to smaller or shifted validation support, and validate selected-case composition prospectively.

Acknowledgments

This work was supported by the National Science Foundation (NSF) under grants 2613950 and 2613761.

References

[1] Ujjwal Baid, Satyam Ghodasara, Suyash Mohan, Michel Bilello, Evan Calabrese, Errol Colak, Keyvan Farahani, Jayashree Kalpathy-Cramer, Felipe C. Kitamura, Sarthak Pati, et al. The RSNA-ASNR-MICCAI BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification, 2021.

[2] Spyridon Bakas, Hamed Akbari, Aristeidis Sotiras, Michel Bilello, Martin Rozycki, Justin S. Kirby, John B. Freymann, Keyvan Farahani, and Christos Davatzikos. Advancing the cancer genome atlas glioma MRI collections with expert segmentation labels and radiomic features. *Scientific Data*, 4: 170117, 2017.

[3] Han Cai, Junyan Li, Muyan Hu, Chuang Gan, and Song Han. EfficientViT: Lightweight multi-scale attention for high-resolution dense prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 17302–17313, 2023.

[4] Wei Feng, Lie Ju, Lin Wang, Kaimin Song, Xin Zhao, and Zongyuan Ge. Unsupervised domain adaptation for medical image segmentation by selective entropy constraints and adaptive semantic alignment. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 623–631, 2023.

[5] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17 (59):1–35, 2016.

[6] Lidia Garrucho, Kaisar Kushibar, C. A. Reidel, Smriti Joshi, Richard Osuala, Apostolia Tsirikoglou, Maciej Bobowicz, Javier del Riego, Alessandro Catanese, Katarzyna Gwoździewicz, et al. A large-scale multicenter breast cancer DCE-MRI benchmark dataset with expert segmentations. *Scientific Data*, 12(1):453, 2025.

[7] Amirata Ghorbani and James Zou. Data shapley: Equitable valuation of data for machine learning. In *Proceedings of the 36th International Conference on Machine Learning*, pages 2242–2251. PMLR, 2019.

[8] Teofilo F. Gonzalez. Clustering to minimize the maximum intercluster distance. *Theoretical Computer Science*, 38:293–306, 1985.

[9] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(25):723–773, 2012.

[10] Hao Guan and Mingxia Liu. Domain adaptation for medical image analysis: A survey. *IEEE Transactions on Biomedical Engineering*, 69(3):1173–1185, 2022.

[11] Nikolaus Hansen, Sibylle D. Müller, and Petros Koumoutsakos. Reducing the time complexity of the derandomized evolution strategy with covariance matrix adaptation (CMA-ES). *Evolutionary Computation*, 11(1):1–18, 2003.

[12] Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021.

[13] Athresh Karanam, Krishnateja Killamsetty, Harsha Kokel, and Rishabh Iyer. ORIENT: Submodular mutual information measures for data subset selection under distribution shift. In *Advances in Neural Information Processing Systems*, pages 31796–31808, 2022.

[14] Krishnateja Killamsetty, Durga Sivasubramanian, Ganesh Ramakrishnan, Abir De, and Rishabh Iyer. Grad-match: Gradient matching based data subset selection for efficient deep model training. In *Proceedings of the 38th International Conference on Machine Learning*, pages 5464–5474. PMLR, 2021.

[15] Krishnateja Killamsetty, Durga Sivasubramanian, Ganesh Ramakrishnan, and Rishabh Iyer. GLISTER: Generalization based data subset selection for efficient and robust learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 8110–8118, 2021.

[16] Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1885–1894. PMLR, 2017.

[17] Stuart Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982.

[18] Mingsheng Long, Yue Cao, Jianmin Wang, and Michael Jordan. Learning transferable features with deep adaptation networks. In *Proceedings of the 32nd International Conference on Machine Learning*, pages 97–105. PMLR, 2015.

[19] Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I. Jordan. Conditional adversarial domain adaptation. In *Advances in Neural Information Processing Systems*, pages 1640–1650, 2018.

[20] Xiaofei Ma, Peng Xu, Zhiguo Wang, Ramesh Nallapati, and Bing Xiang. Domain adaptation with BERT-based domain classification and data selection. In *Proceedings of the 2nd Workshop on Deep Learning Approaches for Low-Resource NLP*, pages 76–83. Association for Computational Linguistics, 2019.

[21] Bjoern H. Menze, Andras Jakab, Stefan Bauer, Jayashree Kalpathy-Cramer, Keyvan Farahani, Justin Kirby, Yuliya Burren, Nicole Porz, Johannes Slotboom, Roland Wiest, et al. The multimodal brain tumor image segmentation benchmark (BRATS). *IEEE Transactions on Medical Imaging*, 34(10): 1993–2024, 2015.

[22] Baharan Mirzasoleiman, Jeff Bilmes, and Jure Leskovec. Coresets for data-efficient training of machine learning models. In *Proceedings of the 37th International Conference on Machine Learning*, pages 6950–6960. PMLR, 2020.

[23] Jialin Peng and Ye Wang. Medical image segmentation with limited supervision: A review of deep network models. *IEEE Access*, 9:36827–36851, 2021.

[24] Xingchao Peng, Qinxun Bai, Xide Xia, Zijun Huang, Kate Saenko, and Bo Wang. Moment matching for multi-source domain adaptation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1406–1415, 2019.

[25] Garima Pruthi, Frederick Liu, Satyen Kale, and Mukund Sundararajan. Estimating training data influence by tracing gradient descent. In *Advances in Neural Information Processing Systems*, pages 19920–19930, 2020.

[26] Mengye Ren, Wenyuan Zeng, Bin Yang, and Raquel Urtasun. Learning to reweight examples for robust deep learning. In *Proceedings of the 35th International Conference on Machine Learning*, pages 4334–4343. PMLR, 2018.

[27] Sebastian Ruder and Barbara Plank. Learning to select data for transfer learning with Bayesian optimization. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 372–382. Association for Computational Linguistics, 2017.

[28] Kuniaki Saito, Donghyun Kim, Stan Sclaroff, Trevor Darrell, and Kate Saenko. Semi-supervised domain adaptation via minimax entropy. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 8050–8058, 2019.

[29] Ozan Sener and Silvio Savarese. Active learning for convolutional neural networks: A core-set approach. In *International Conference on Learning Representations*, 2018.

[30] Jun Shu, Qi Xie, Lixuan Yi, Qian Zhao, Sanping Zhou, Zongben Xu, and Deyu Meng. Meta-Weight-Net: Learning an explicit mapping for sample weighting. In *Advances in Neural Information Processing Systems*, pages 1919–1930, 2019.

[31] Ankit Singh. CLDA: Contrastive learning for semi-supervised domain adaptation. In *Advances in Neural Information Processing Systems*, pages 5089–5101, 2021.

[32] Baochen Sun and Kate Saenko. Deep CORAL: Correlation alignment for deep domain adaptation. In *Computer Vision – ECCV 2016 Workshops*, pages 443–450. Springer, 2016.

[33] Hemanth Venkateswara, Jose Eusebio, Shayok Chakraborty, and Sethuraman Panchanathan. Deep hashing network for unsupervised domain adaptation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 5018–5027, 2017.

[34] Tuan-Hung Vu, Himalaya Jain, Maxime Bucher, Matthieu Cord, and Patrick Perez. ADVENT: Adversarial entropy minimization for domain adaptation in semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2517–2526, 2019.

[35] Xinyi Wang, Hieu Pham, Paul Michel, Antonios Anastasopoulos, Jaime Carbonell, and Graham Neubig. Optimizing data usage via differentiable rewards. In *Proceedings of the 37th International Conference on Machine Learning*, pages 9983–9995. PMLR, 2020.

[36] Kai Wei, Rishabh Iyer, and Jeff Bilmes. Submodularity in data subset selection and active learning. In *Proceedings of the 32nd International Conference on Machine Learning*, pages 1954–1963. PMLR, 2015.

[37] Mengzhou Xia, Sadhika Malladi, Suchin Gururangan, Sanjeev Arora, and Danqi Chen. LESS: Selecting influential data for targeted instruction tuning. In *Proceedings of the 41st International Conference on Machine Learning*, pages 54104–54132. PMLR, 2024.

[38] Jinsung Yoon, Sercan Arik, and Tomas Pfister. Data valuation using reinforcement learning. In *Proceedings of the 37th International Conference on Machine Learning*, pages 10842–10851. PMLR, 2020.

[39] Jee Seok Yoon, Kwanseok Oh, Yooseung Shin, Maciej A. Mazurowski, and Heung-Il Suk. Domain generalization for medical image analysis: A review. *Proceedings of the IEEE*, 112(10):1583–1609, 2024.

[40] Wen Zhang, Lingfei Deng, Lei Zhang, and Dongrui Wu. A survey on negative transfer. *IEEE/CAA Journal of Automatica Sinica*, 10(2):305–329, 2023.

[41] Yihua Zhang, Yimeng Zhang, Aochuan Chen, Jinghan Jia, Jiancheng Liu, Gaowen Liu, Mingyi Hong, Shiyu Chang, and Sijia Liu. Selectivity drives productivity: Efficient dataset pruning for enhanced transfer learning. In *Advances in Neural Information Processing Systems*, pages 36913–36937, 2023.

[42] Han Zhao, Shanghang Zhang, Guanhang Wu, João P. Costeira, José M. F. Moura, and Geoffrey J. Gordon. Adversarial multiple source domain adaptation. In *Advances in Neural Information Processing Systems*, pages 8559–8570, 2018.

[43] Bolei Zhou, Hang Zhao, Xavier Puig, Sanja Fidler, Adela Barriuso, and Antonio Torralba. Scene parsing through ADE20K dataset. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 633–641, 2017.